

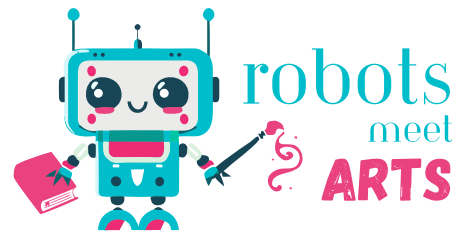


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EXPLORING HISTORY



**Subject:**

History, Computational Thinking

Content:

- European explorers and their routes
- Decomposition and sequencing of tasks
- Using precise instructions to give commands

**Computational practices:
choose what fits**

- Decomposing
- Pattern recognition
- Abstraction
- Algorithms
- Problem solving

Unplugged**Preparation time:**

10 minutes

**Age:**

8-12 years

**Duration:**

60 minutes

**Location:**

Classroom, art room

**Group distribution:**

Pairs or small groups (2-4 students)

**Difficulty****After this lesson, the children will be able to...**

- Identify the journeys of different European explorers.
- Understand the importance of clear, sequenced instructions.
- Relate algorithmic thinking to historical navigation.
- Collaborate to solve challenges and follow commands.
- Develop critical thinking and problem-solving skills.
- Apply spatial awareness and direction-following strategies.



Materials needed:

- Printed maps of explorers' routes
- Instruction cards (move forward X steps, turn left, turn right, stop, etc.)
- Small tokens or cutouts representing ships or explorers
- Pencils and A4 paper

Preparation and teacher notes

- Print and prepare historical maps with the routes of different explorers.
- Ensure students have access to the instruction cards and ship tokens.
- Discuss key vocabulary beforehand (e.g., navigation, sequence, debugging).
- Encourage students to think step-by-step when creating their algorithms.
- Remind students that making mistakes is part of learning—debugging is key

Warm up 15'

1. Introduction:

- Project a historical map and ask guiding questions:
 - Who were the explorers? What were they searching for?
 - How do you think they navigated without modern technology?
 - Why is it important to give clear and structured instructions?

2. Teacher Demonstration:

- The teacher follows incorrect or unclear student instructions (e.g., "Turn right" but without specifying how far).
- This creates errors, helping students realize the importance of precision in instructions.

Preparation and teacher notes

Each group will receive the following materials.

- Printed maps of explorers' routes
- Instruction cards (move forward X steps, turn left, turn right, stop, etc.)
- Small tokens or cutouts representing ships and explorers
- Pencils and A4 paper

Differentiation:

- To make oral instructions easier, have the children work with geometric shapes first.
- Give the artist a max number of instruction it can use.
- Have the artist write down the instructions on paper.
- Use organic shapes instead of known geometric shapes.
- Use more/less figures to make the task easier or more difficult

Build up 20'

Programming the Explorer

- Each group receives a map and an assigned explorer (e.g., Columbus, Magellan).
- They create a sequence of instructions to recreate the explorer's journey.
- One student acts as the "Navigator" and follows the group's commands, moving their token on the map.

Rehearsal 25'

Debugging the Algorithm

- Other groups review the given instructions to check if the route was followed correctly.
- If the "explorer" gets lost or doesn't reach the destination, the group refines the instructions.
- The process is repeated until the route is correctly navigated.

Clean up and reflection 15'

The children clean up, order the materials and place them in the right spot.

Finally, reflection:

- Groups compare their journeys and discuss their strategies.
- Discussion prompts:
 - What problems did you encounter while giving instructions?
 - Why do you think sequencing and precision matter?
 - How does this activity relate to computer programming?